SCU | Southern California University of Health Sciences

PERIPHERAL NERVOUS SYSTEM

Module 2

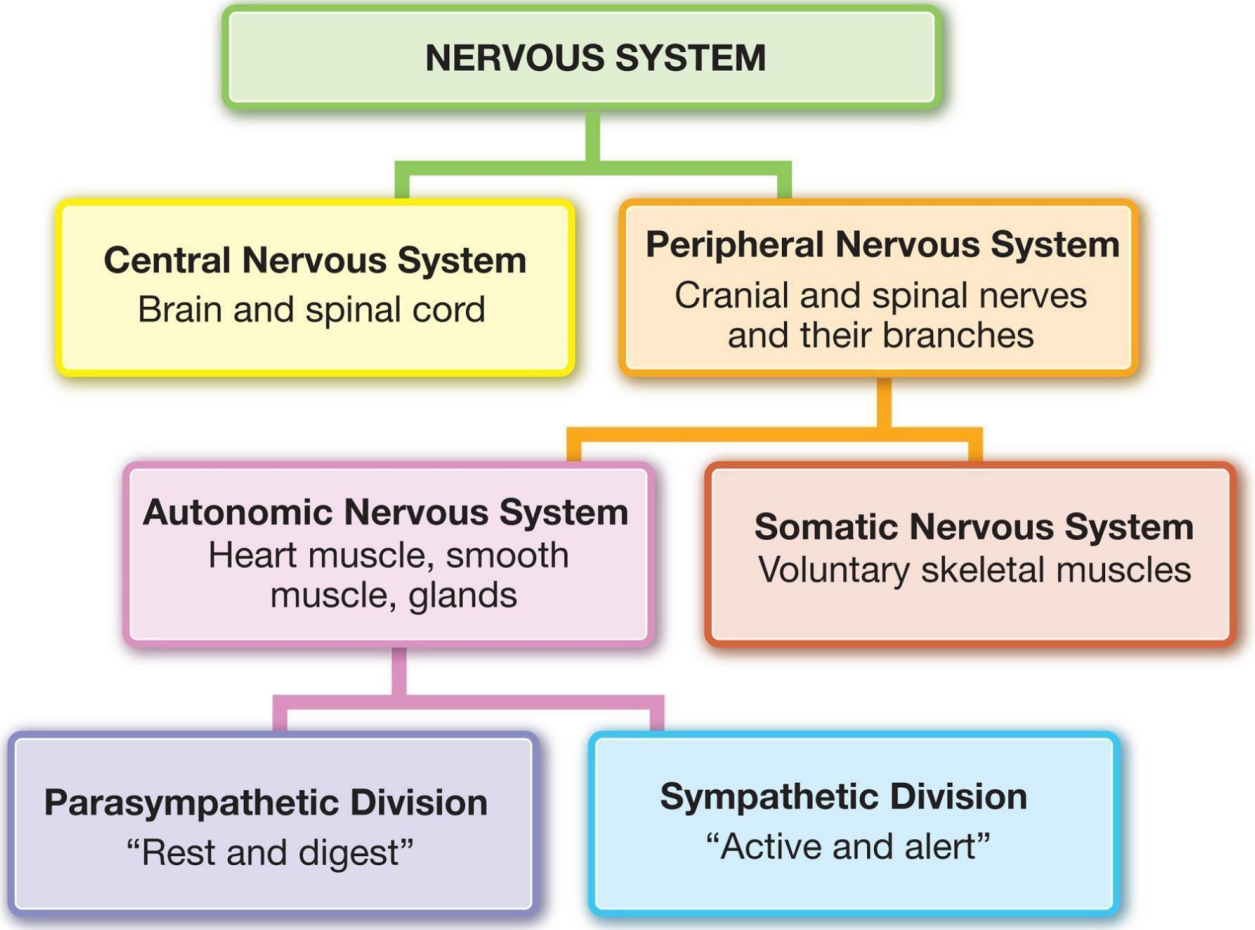
READING ASSIGNMENT

- Moore's Clinically Oriented Anatomy-The Nervous System

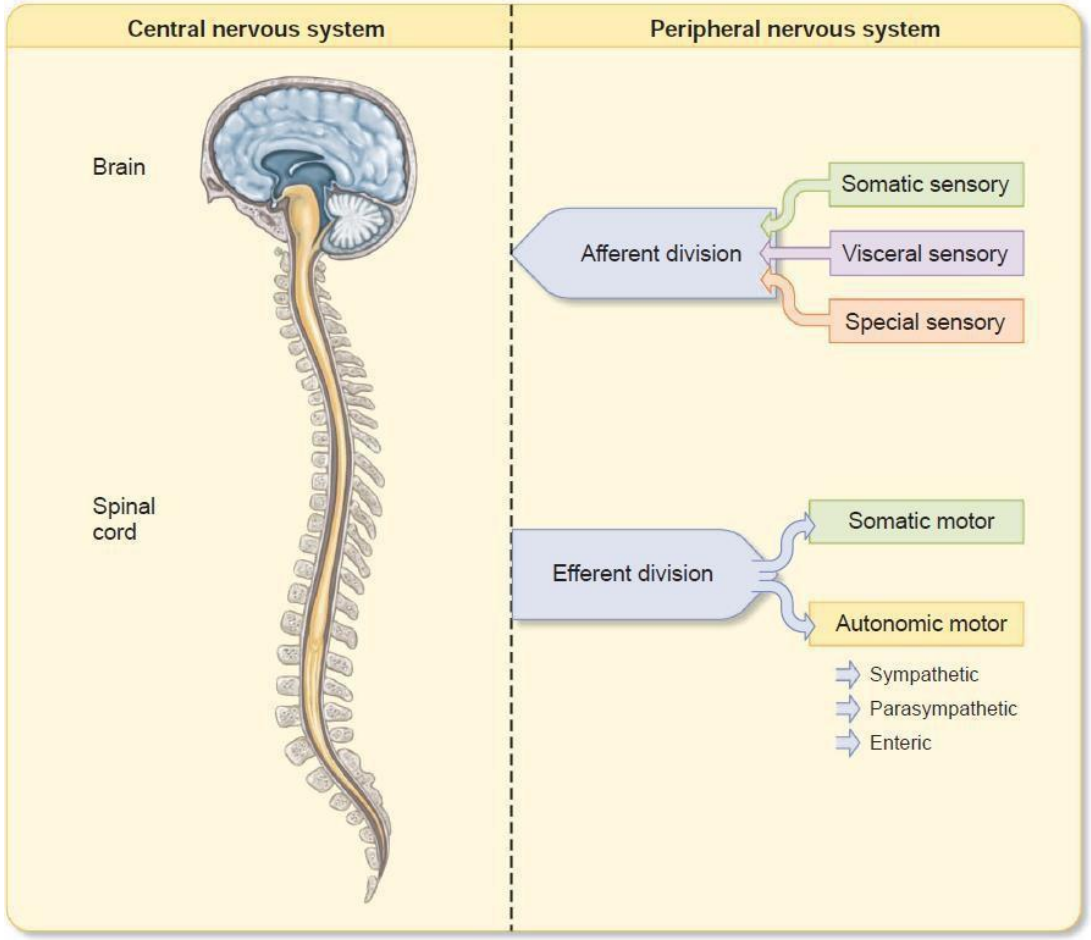
LEARNING OBJECTIVES

1. Describe the structural and functional divisions of the Nervous System.
2. Briefly describe the structure of a neuron.
3. List the components of the central and peripheral nervous systems.
4. Briefly describe the structure of the spinal cord in a cross section.
5. Indicate the number of spinal nerves in each region of the spine.
6. Describe the formation, branches and types of fibers (functional components) of a typical spinal nerve.
7. Name the plexuses formed by the ventral rami of spinal nerves, the nerves that form each plexus and the territory they supply.
8. Compare the efferent components of the somatic and autonomic nervous systems.
9. Compare the sympathetic and parasympathetic divisions of the autonomic nervous system in terms of their structure and function.
10. Define dermatome and give examples of cervical, thoracic, lumbar and sacral dermatomes.

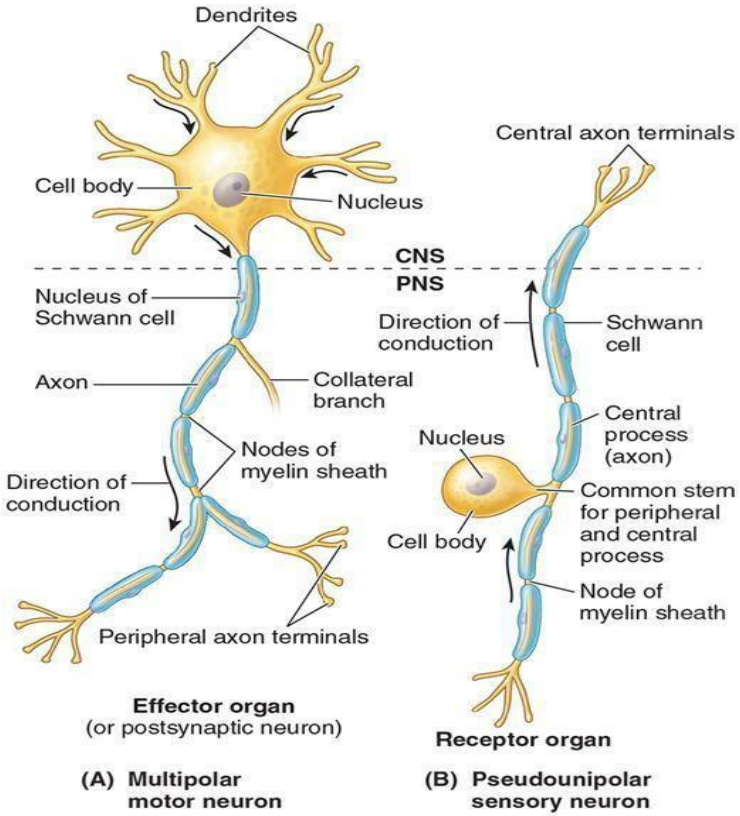
ORGANIZATION OIF THE NERVOUS SYSTEM

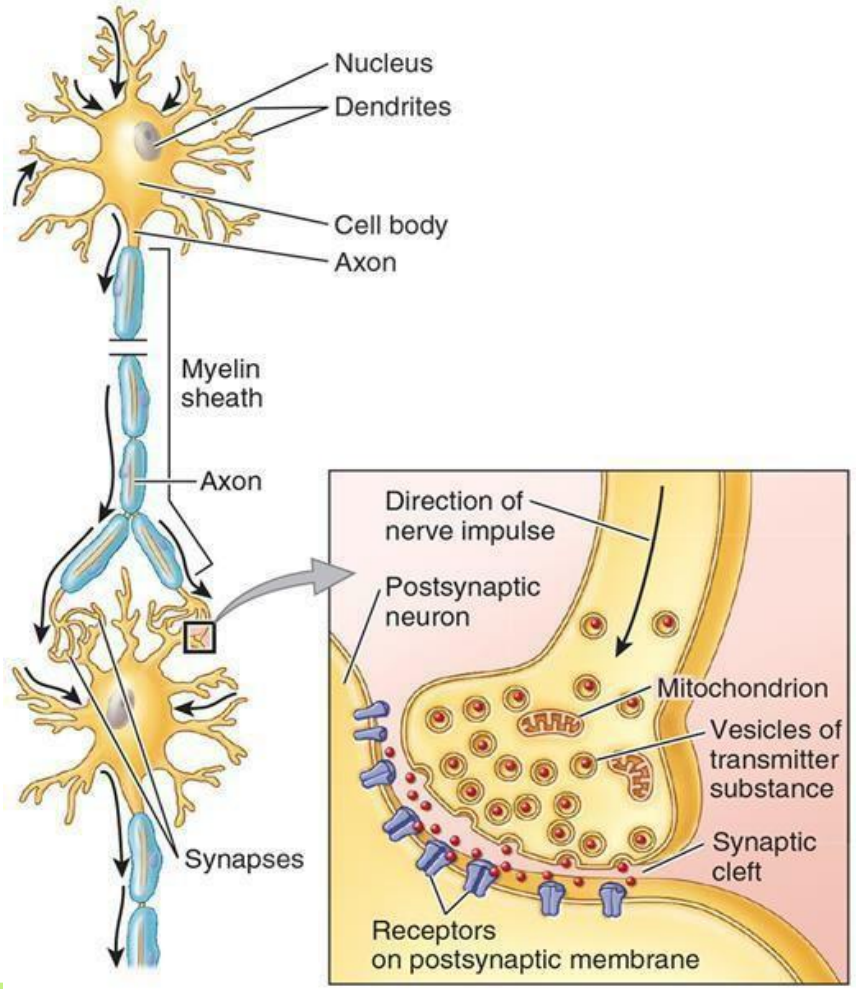


DIVISIONS OF THE NERVOUS SYSTEM



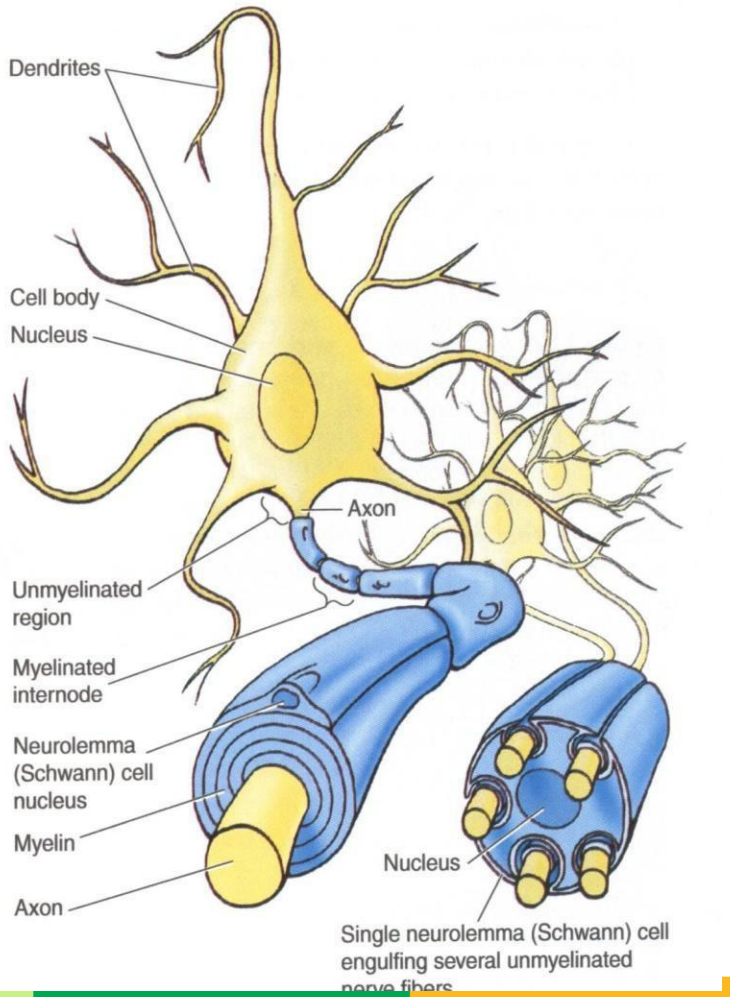
ORGANIZATION OF NERVOUS SYSTEM





PERIPHERAL NERVOUS SYSTEM

- Peripheral nerve fibers
  - Myelinated
    - Schwann cells
  - Unmyelinated
- Ganglia (ganglion)
  - Spinal ganglia
  - Autonomic ganglia
    - Sympathetic ganglia
    - Parasympathetic ganglia

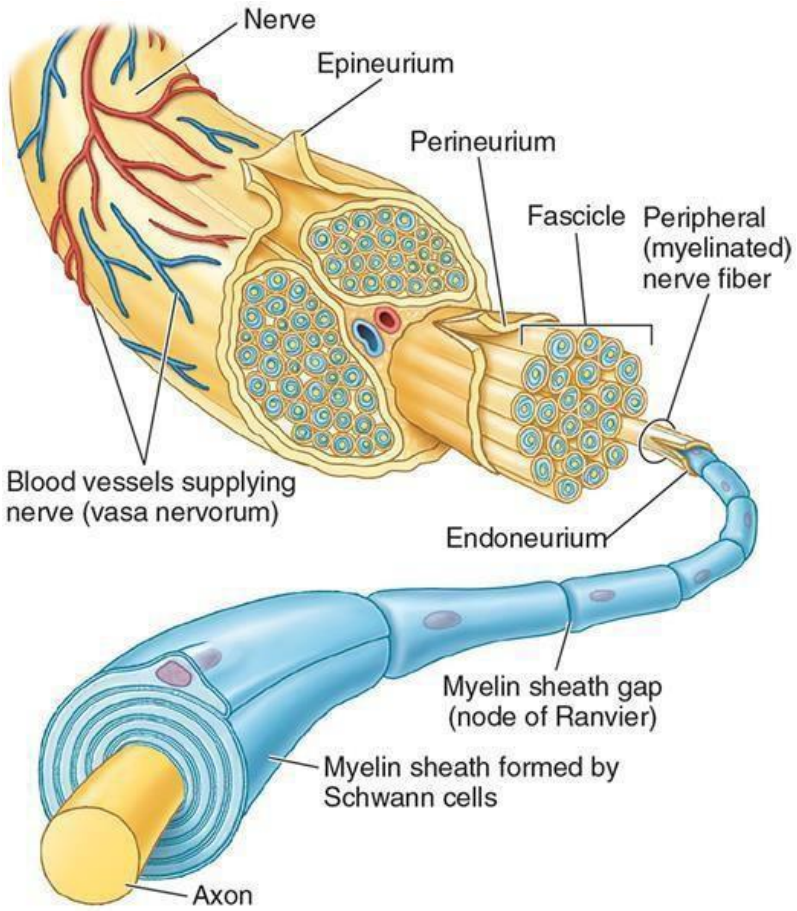


THE ANATOMY OF A NERVE

A **nerve** consists of the following components:

- A bundle of nerve fibers outside the CNS (or a "bundle of bundled fibers," or *fascicles*, in the case of a larger nerve)

- The connective tissue coverings that surround and bind the nerve fibers and fascicles together
- The blood vessels (*vasa nervorum*) that nourish the nerve fibers and their coverings



CRANIAL AND SPINAL NERVES

Nerves are either cranial nerves or spinal nerves, or derivatives of them:

- **Cranial nerves** exit the cranial cavity through foramina (openings) in the cranium and are identified by a descriptive name (e.g., “trochlear nerve”) or a Roman numeral (e.g., “CN IV”). Only 11 of the 12 pairs of cranial nerves arise from the brain; the other pair (CN XI) arises from the superior part of the spinal cord.
- **Spinal (segmental) nerves** exit the vertebral column (spine) through intervertebral foramina. Spinal nerves arise in bilateral pairs from a specific segment of the spinal cord. The 31 spinal cord segments and the 31 pairs of nerves arising from them are identified by a letter and number (e.g., “T4”) designating the region of the spinal cord and their superior-to-inferior order (C, cervical; T, thoracic; L, lumbar; S, sacral; Co, coccygeal).

PERIPHERAL NERVES

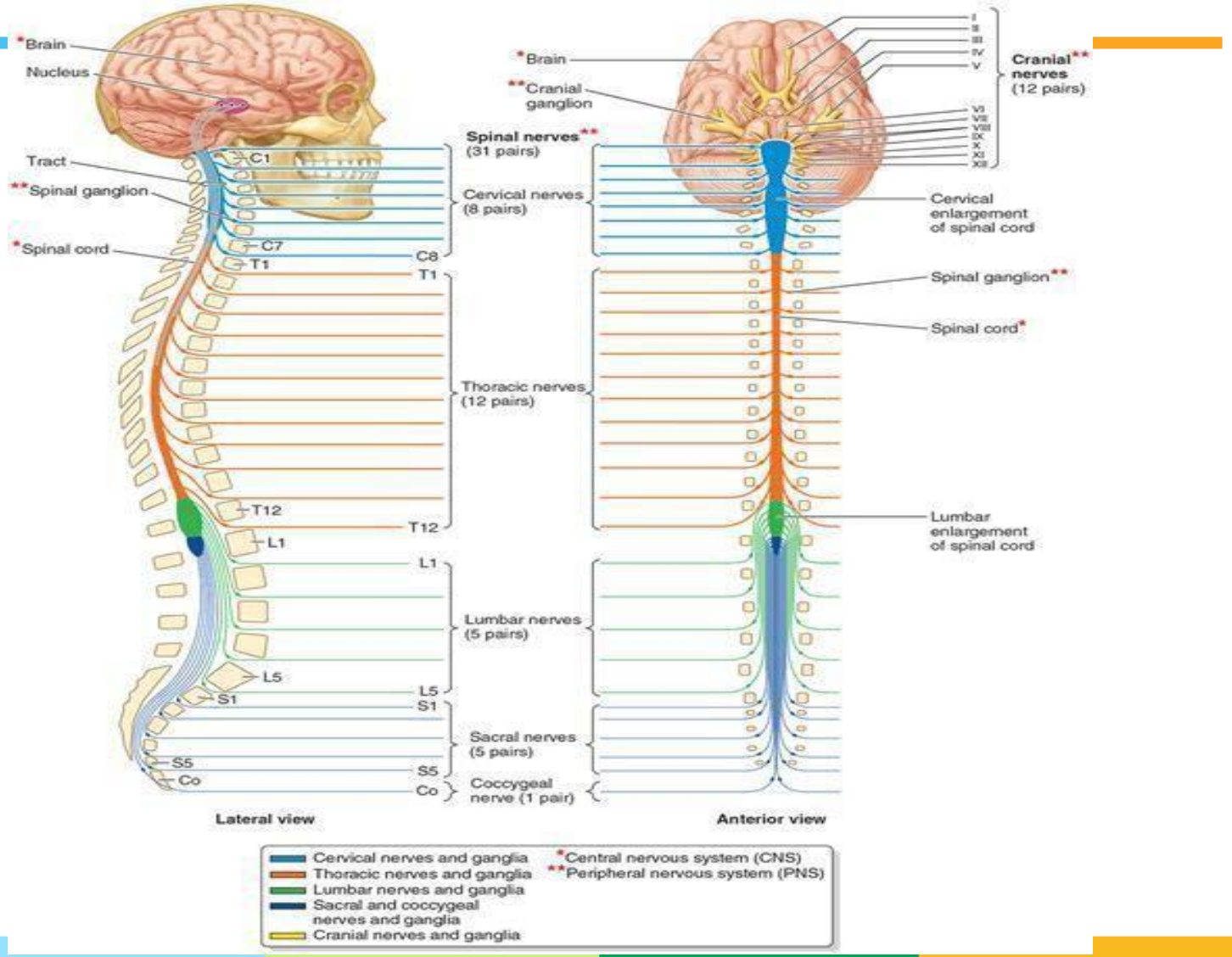
Cranial nerves

1. Olfactory
2. Optic
3. Oculomotor
4. Trochlear
5. Trigeminal
6. Abducens
7. Facial
8. Vestibulocochlear
9. Glossopharyngeal
10. Vagus
11. Spinal accessory
12. Hypoglossal

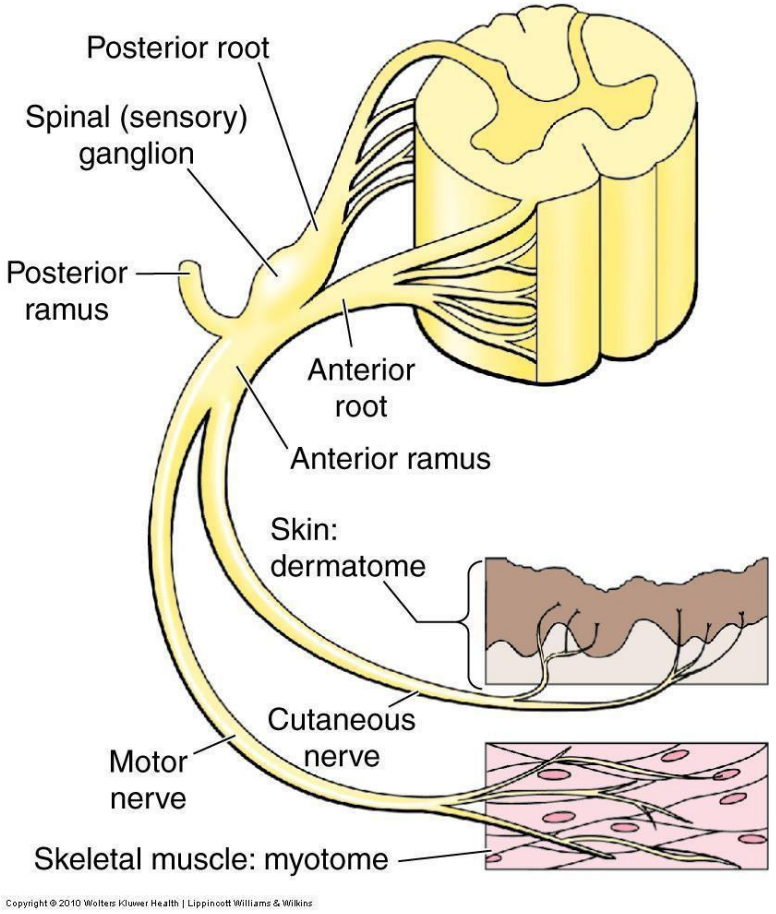
Spinal nerves

- Cervical (8)
- Thoracic (12)
- Lumbar (5)
- Sacral (5)
- Coccygeal (1)

ORGANIZATION OF PERIPHERAL NERVOUS SYSTEM



A SPINAL NERVE



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- Spinal nerves initially arise from the spinal cord as **rootlet**; the rootlets converge to form two nerve roots.

An anterior (ventral) nerve root, consisting of motor (efferent) fibers passing from nerve cell bodies in the anterior and lateral horns of spinal cord gray matter to effector organs located peripherally

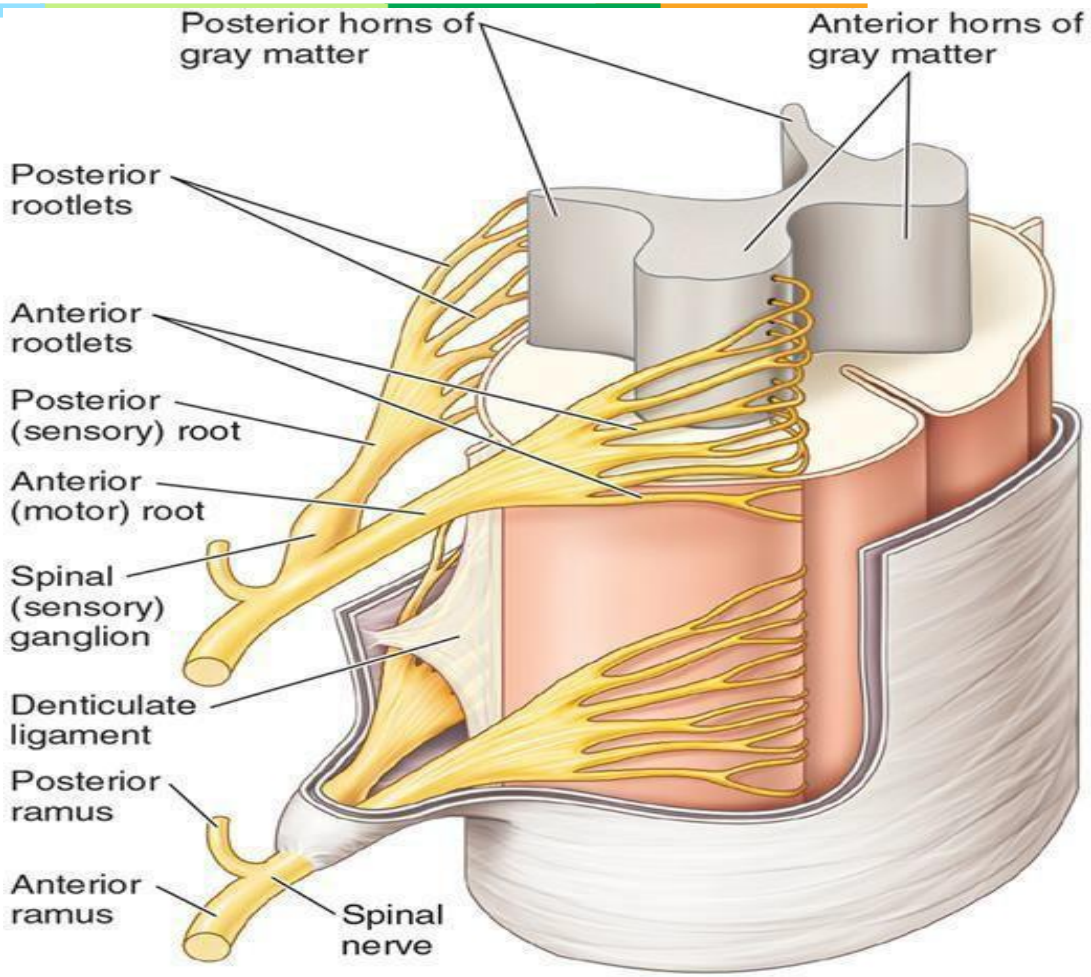
A **posterior (dorsal) nerve root**, consisting of sensory (afferent) fibers from cell bodies in the spinal (sensory) or posterior (dorsal) root ganglion (commonly abbreviated in clinical use as “DRG”) that extend peripherally to sensory endings and centrally to the posterior horn of spinal cord gray matter

A SPINAL NERVE

The posterior and anterior nerve roots unite, within or just proximal to the intervertebral foramen, to form a mixed (both motor and sensory) *spinal nerve*, which immediately divides into

- a ***posterior (dorsal) ramus*** and
- an ***anterior (ventral) ramus***.

As branches of the mixed spinal nerve, the posterior and anterior rami carry both motor and sensory fibers, as do all their subsequent branches.



DORSAL AND VENTRAL RAMI

1. **Posterior (Dorsal) rami of spinal nerves** supply nerve fibers to the synovial joints of the vertebral column, deep (epaxial) muscles of the back, and the overlying skin in a segmental pattern. (As a general rule, the posterior rami remain separate from each other (do not merge to form major somatic nerve plexuses).

2. **Anterior (Ventral) rami of spinal nerves** supply nerve fibers to the much larger remaining area, consisting of the skin and muscles of the anterior and lateral regions of the trunk and the upper and lower limbs.

COMPONENTS OF A TYPICAL SPINAL NERVE

Somatic fibers

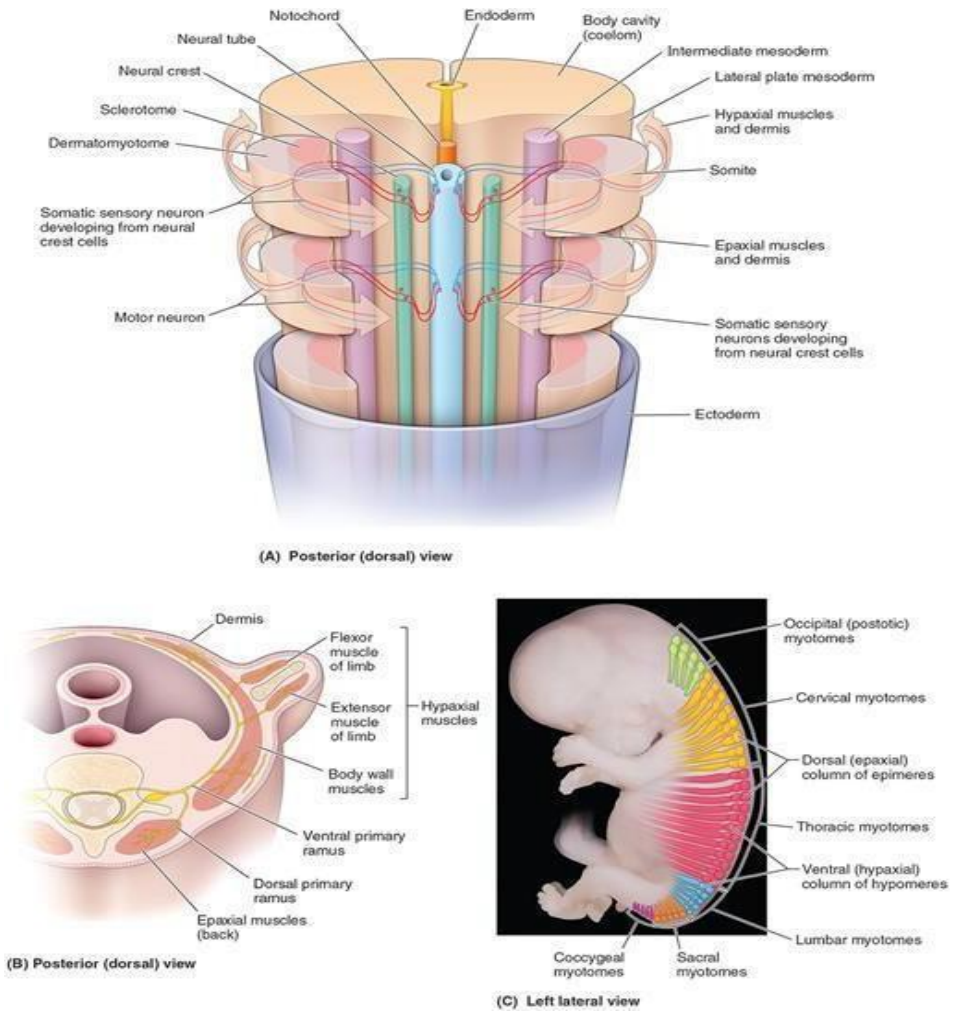
- Sensory (general somatic afferent =GSA)
- Motor (general somatic efferent =GSE)

Visceral fibers

- Sensory (general visceral afferent =GVA)
- Motor (general visceral efferent = GVE)
  - Presynaptic or preganglionic fiber
  - Postsynaptic or postganglionic fiber

DERMATOMES-EMBRYOLOGICAL BASIS

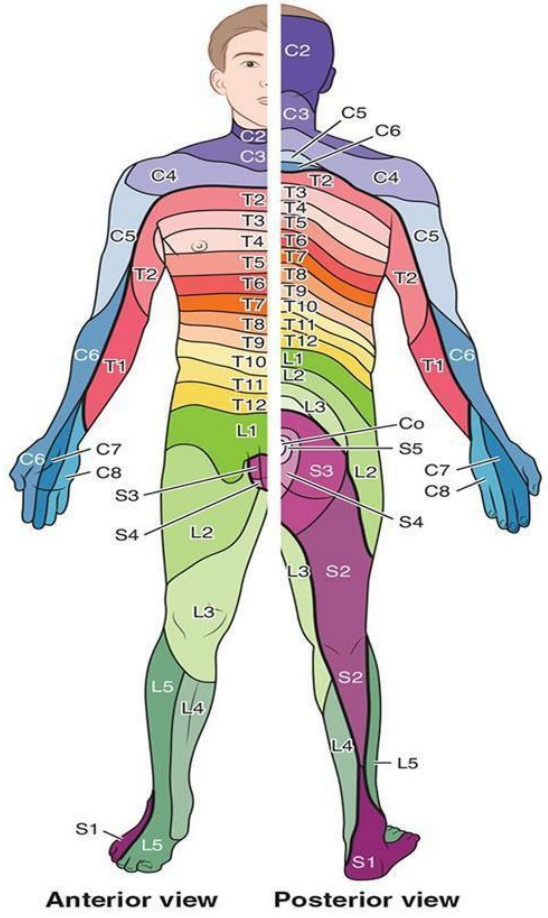
- The relationship between the nerves and the tissue derived from the dermatomyotome remains throughout life:
- The unilateral area of skin supplied by a single (right or left member of a pair of) spinal nerves is called a **dermatome**.
- The unilateral mass of muscle supplied by a single spinal nerve is called a **myotome**.



Dermatomes (segmental cutaneous innervation

Dermatome maps of the body are based on an accumulation of clinical findings following spinal nerve injuries. This map reflects both anatomical (actual) distribution or segmental innervation and clinical experience.

Spinal nerve C1 lacks a significant afferent component and does not supply the skin; therefore, no C1 dermatome is depicted. Note that in the Foerster map, C5–T1 and L3–S1 are distributed almost entirely in the limbs (i.e., have little or no representation on the trunk).



SOME IMPORTANT DERMATOMES TO REMEMBER

C 2 – back of the head; the “highest” dermatome of the body

C 4 – top part of the shoulder

C 6 – thumb

C7 – middle finger C 8 – little finger

T 4 – level of the nipples

T 10 – level of the umbilicus.

T 12/L 1 – the skin along the inguinal crease L 4 – big toe

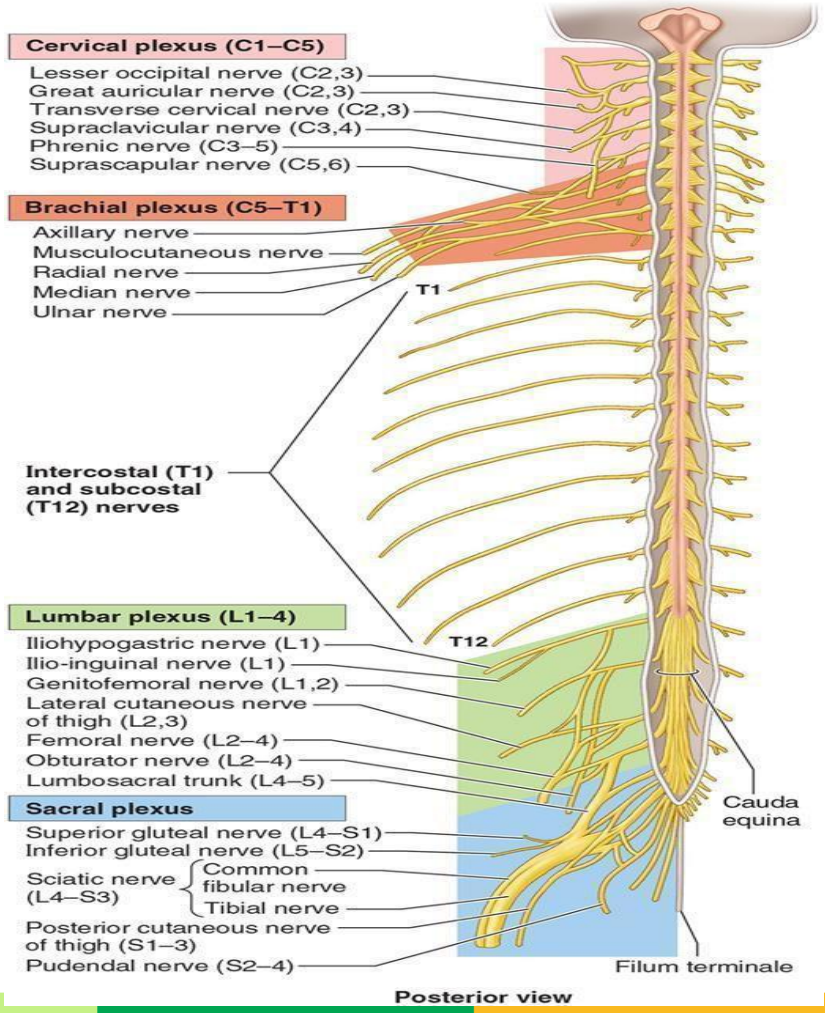
S 1 – little toe & sole of the foot

S 4/S 5/Cocc 1 – the skin around the anus

THE SOMATIC PLEXUSES

The majority of anterior rami merge with one or more adjacent anterior rami, forming the major *somatic nerve plexuses* (networks) in which their fibers intermingle and from which a new set of ***multisegmental*** *peripheral nerves* emerges.

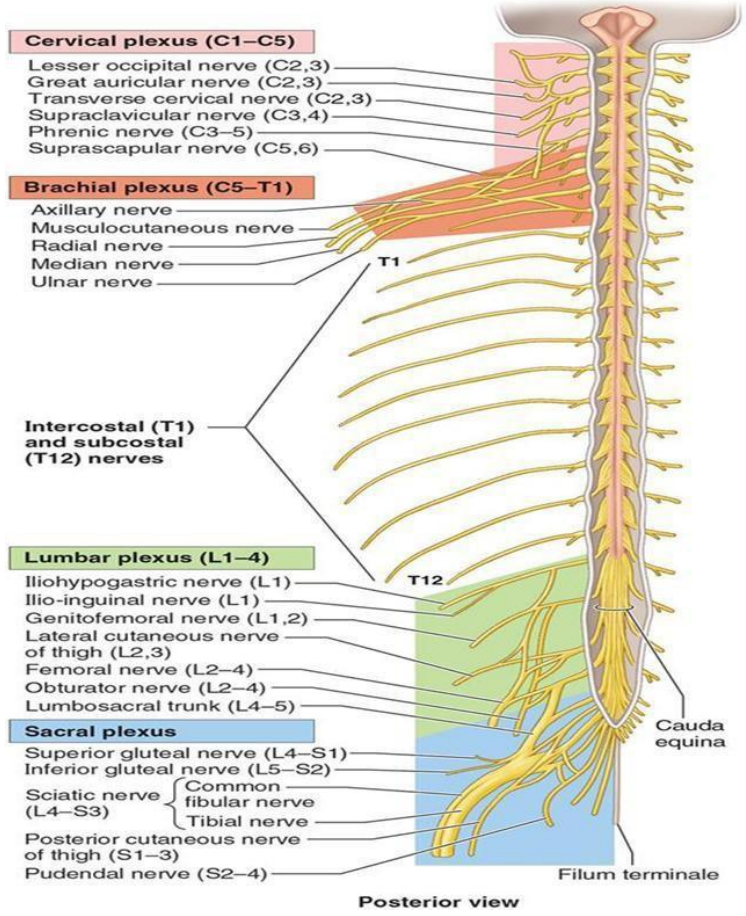
The **anterior rami** of spinal nerves participating in plexus formation contribute fibers to multiple peripheral nerves arising from the plexus.



SOMATIC NERVE PLEXUSES

It is a complex interwoven network of nerves formed by ventral rami of adjacent spinal nerves.

- Cervical plexus (C1-C4)
- Brachial plexus (C5-T1)
- Lumbar plexus (L1-L4)
- Sacral plexus (L4-S4)
- Coccygeal plexus (S4-Co1)



CRANIAL NERVES

Names of the Cranial Nerves.

CN I ------ OLFACTORY NERVE

CN II ----- OPTIC NERVE

CN III ---- OCULOMOTOR NERVE

CN IV --- TROCHLEAR NERVE

CN V ---- TRIGEMINAL NERVE

CN VI --- ABDUCENS NERVE

CN VII -- FACIAL NERVE

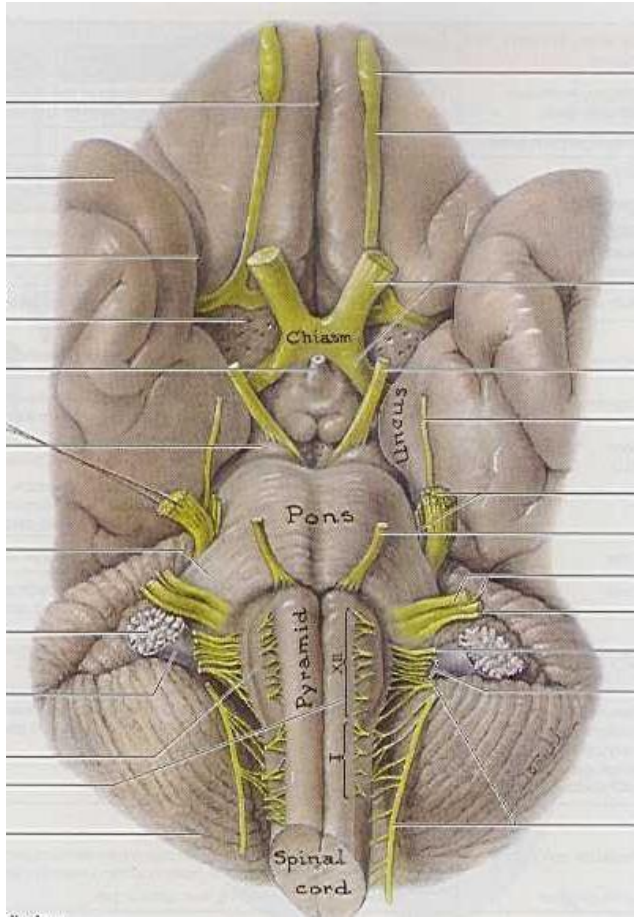
CN VIII - AUDITORY (VESTIBULOCOCHLEAR) NERVE

CN IX --- GLOSSOPHARYNGEAL NERVE

CN X ---- VAGUS NERVE

CN XI --- SPINAL ACCESSORY NERVE

CN XII -- HYPOGLOSSAL NERVE



COMPONENTS OF A CRANIAL NERVE

Somatic fibers

- Sensory (general somatic afferent =GSA)
- Motor (general somatic efferent =GSE)

Visceral fibers

- Sensory (general visceral afferent =GVA)
- Motor (general visceral efferent = GVE)
  - Presynaptic or preganglionic fiber
  - Postsynaptic or postganglionic fiber

Special fibers

- Special somatic afferent =SSA
- Special visceral afferent =SVA
- Special visceral efferent =SVE (branchial motor)

FUNCTIONAL TYPE OF NERVE FIBERS

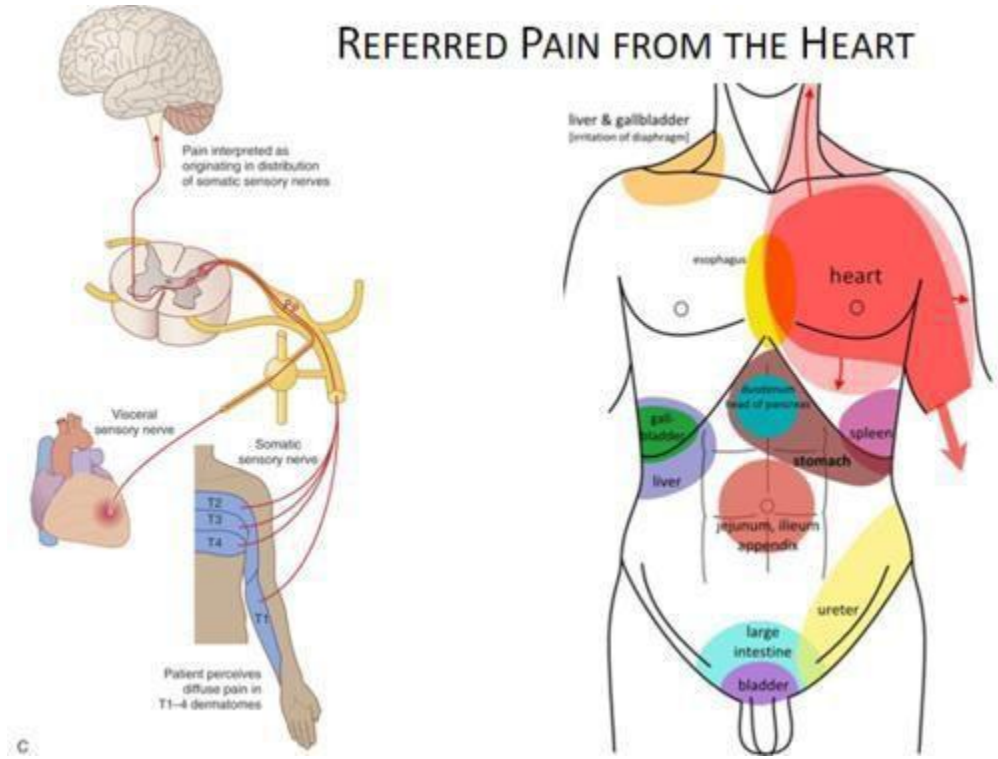
- GSE fibers send motor stimuli to all skeletal muscles, except the skeletal muscles derived from pharyngeal arches.
- SVE fibers send motor stimuli to all skeletal muscles derived from pharyngeal arches and these muscles are:
  - Muscles of facial expression (CN VII)
  - Muscles of mastication(chewing) (CN V)
  - Pharyngeal muscles (swallowing) (CN IX, X)
  - Laryngeal muscles (phonation=voice production) (CN X)
  - Trapezius & sternocleidomastoid muscles (CN XI)
- GVE fibers send motor stimuli to smooth muscles, cardiac muscle, and glandular tissues.
- GSA fibers receive sensory stimuli coming from somatic structures such as skin, superficial and deep fasciae, skeletal muscles (pain & proprioceptive sensations), bones, joints, ligaments, and tendons.
- GVA fibers receive sensory stimuli coming from visceral organs.
- SSA fibers receive visual and auditory stimuli through CN II and VIII.
- SVA fibers receive olfactory (smell) stimuli through CN I and gustatory (taste) stimuli through CN VII, IX, & X.

REFERRED PAIN

Referred pain occurs when sensory information comes to the spinal cord, but is interpreted by the CNS as coming from another location innervated by the same spinal cord level

Types:

- Viscero-somatic
- Somato-somatic



AUTONOMIC NERVOUS SYSTEM

- Carries fibers to autonomic effectors
- Major function – to regulate heartbeat, smooth muscle contraction, and glandular secretions to maintain homeostasis
- Two efferent divisions
  - Sympathetic
  - Parasympathetic
- Many autonomic effectors are dually innervated, which allows remarkably precise control of effector

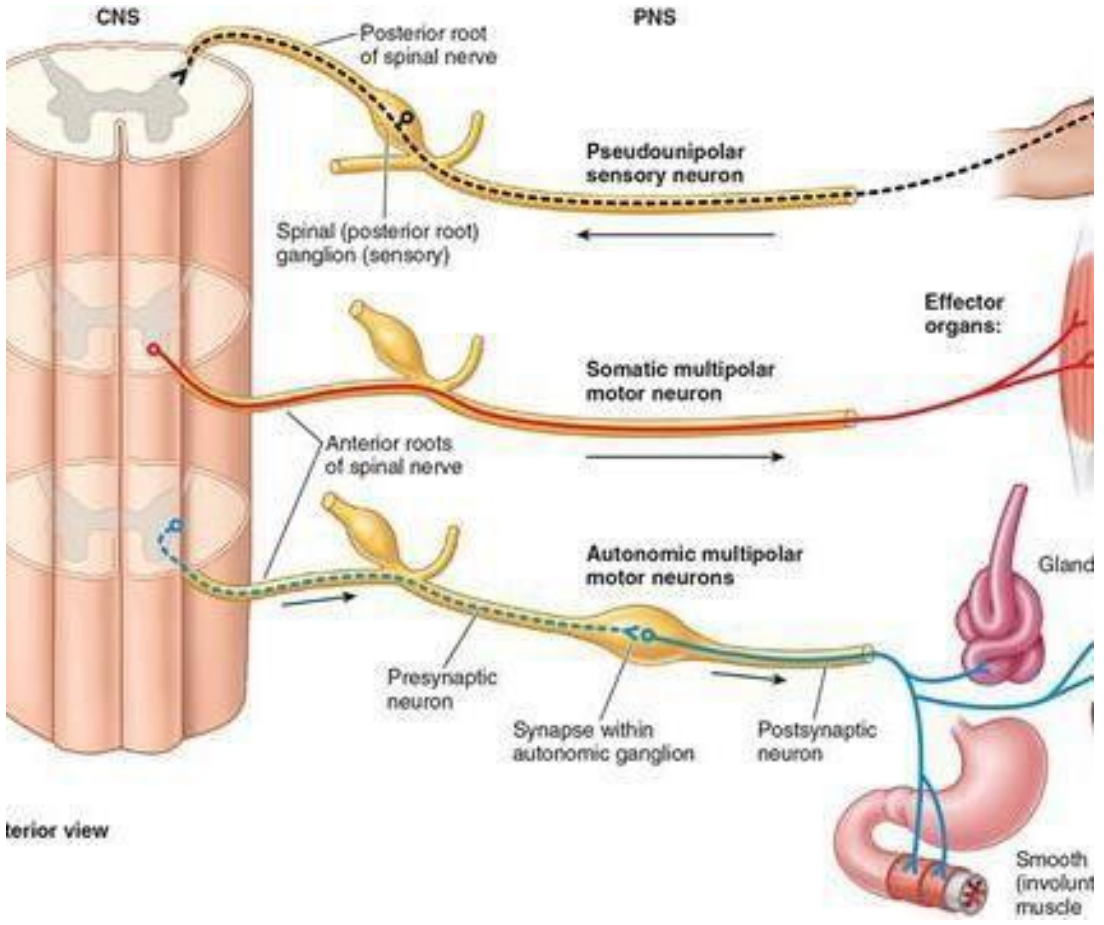
ORGANIZATION OF AUTONOMIC NERVOUS SYSTEM

- In both divisions of the ANS, conduction of impulses from the CNS to the effector organ involves a series of two multipolar neurons .

1. **Preganglionic neuron**
2. **Postganglionic neuron**

ORGANIZATION IOF AUTONOMIC NERVOUS SYSTEM

- The nerve cell body of the first **presynaptic (preganglionic) neuron** is located in the gray matter of the CNS (brain and spinal cord).
- Its fiber (axon) synapses only on the cell body of a **postsynaptic (postganglionic) neuron**, the second neuron in the series.
- The cell bodies of these second neurons are located outside the CNS in autonomic ganglia, with fibers terminating on the effector organ (smooth muscle, modified cardiac muscle, or glands).

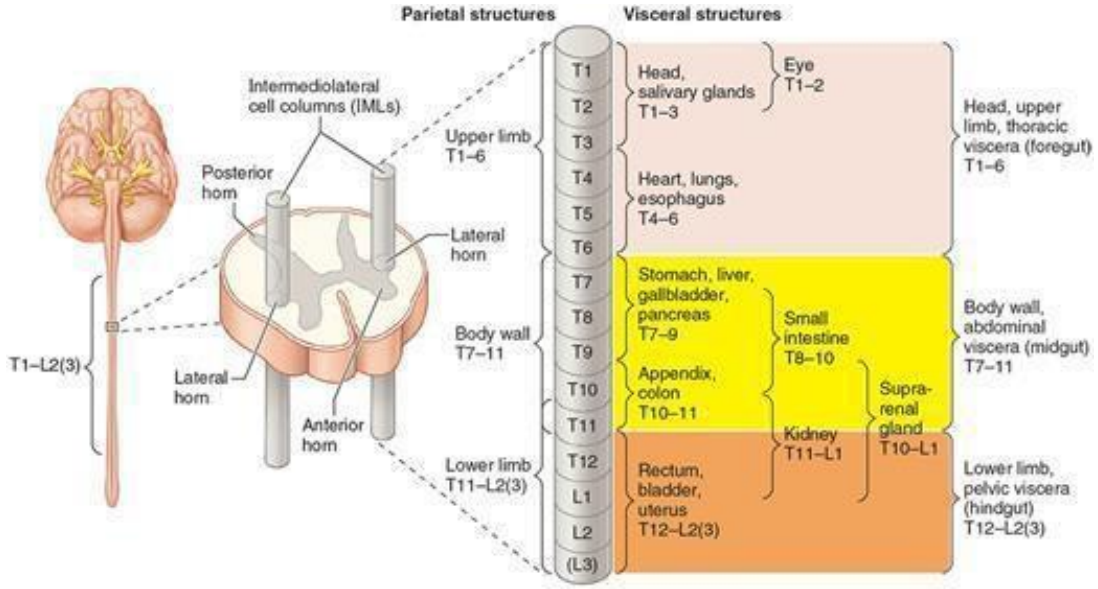


AUTONOMIC NERVOUS SYSTEM

Sympathetic nervous system	Parasympathetic nervous system
Thoracolumbar origin	Craniosacral origin (brainstem, S2-S4)
Types of SP ganglia: • Paravertebral • Prevertebral	Types of PS ganglia: • Cranial • Intramural
Energy-expending (“fight, flight, or fright”)	Energy-conserving (“rest and digest”)

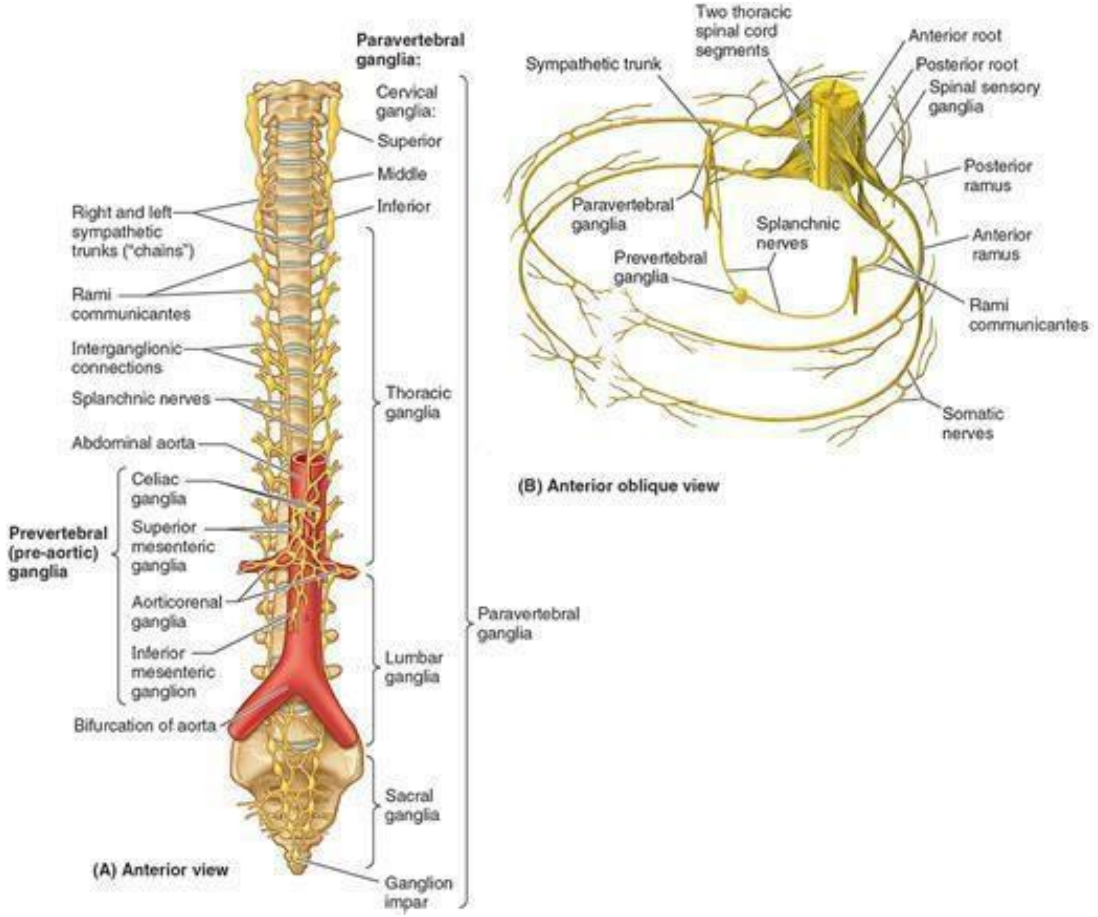
SYMPATHETIC DIVISION OF ANS

- The cell bodies of the **presynaptic neurons** of the sympathetic division of the ANS are found in only one location: the **intermediolateral cell columns** (IMLs) or lateral horn in the thoracic and the upper lumbar (T1–L2 or 3) segments of the spinal cord.(hence the alternate name "thoracolumbar division).



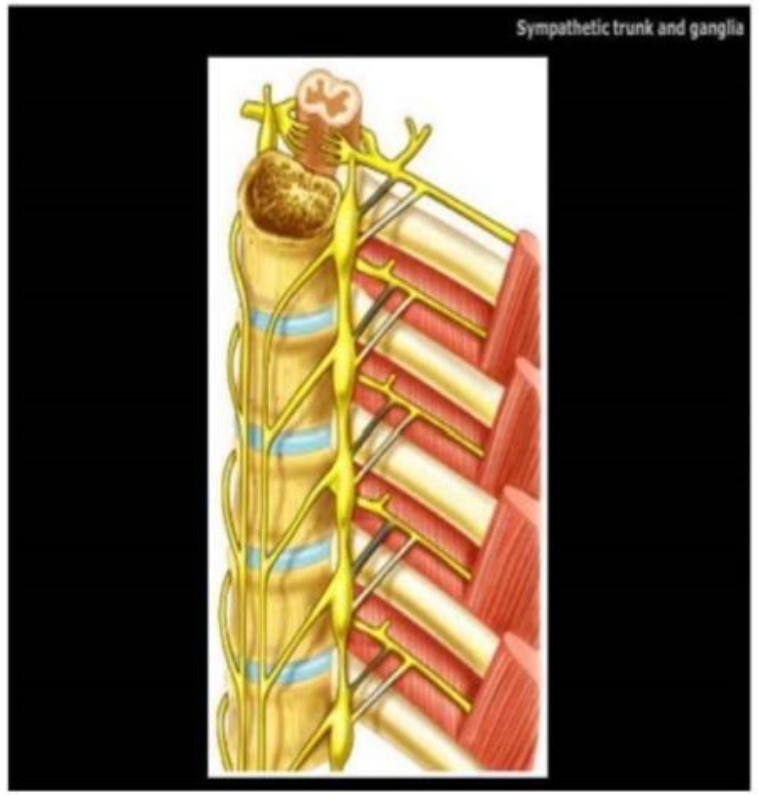
SYMPATHETIC NERVOUS SYSTEM

The cell bodies of **postsynaptic neurons** of the sympathetic nervous system occur in two locations, **the paravertebral and prevertebral ganglia**.



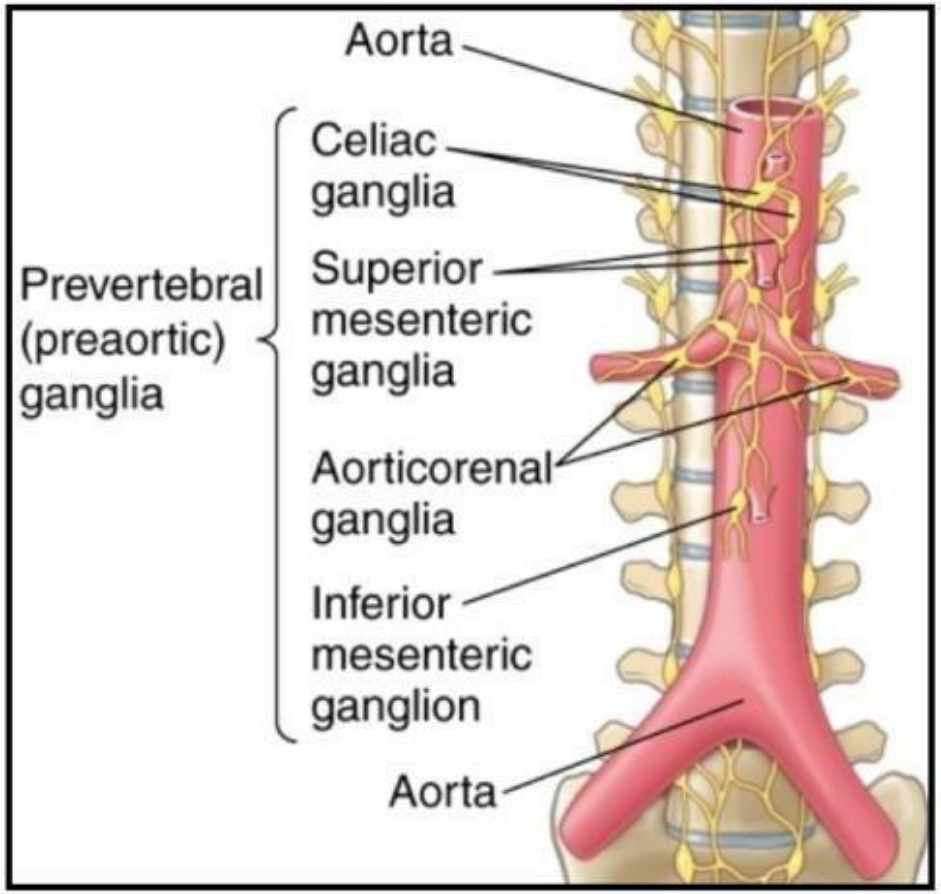
PARAVERTEBRAL GANGLIA

- **Paravertebral ganglia** are linked to form right and left *sympathetic trunks* (chains) on each side of the vertebral column and extend essentially the length of this column. The superior paravertebral ganglion (the *superior cervical ganglion* of each sympathetic trunk) lies at the base of the cranium. The *ganglion impar* forms inferiorly where the two trunks unite at the level of the coccyx.

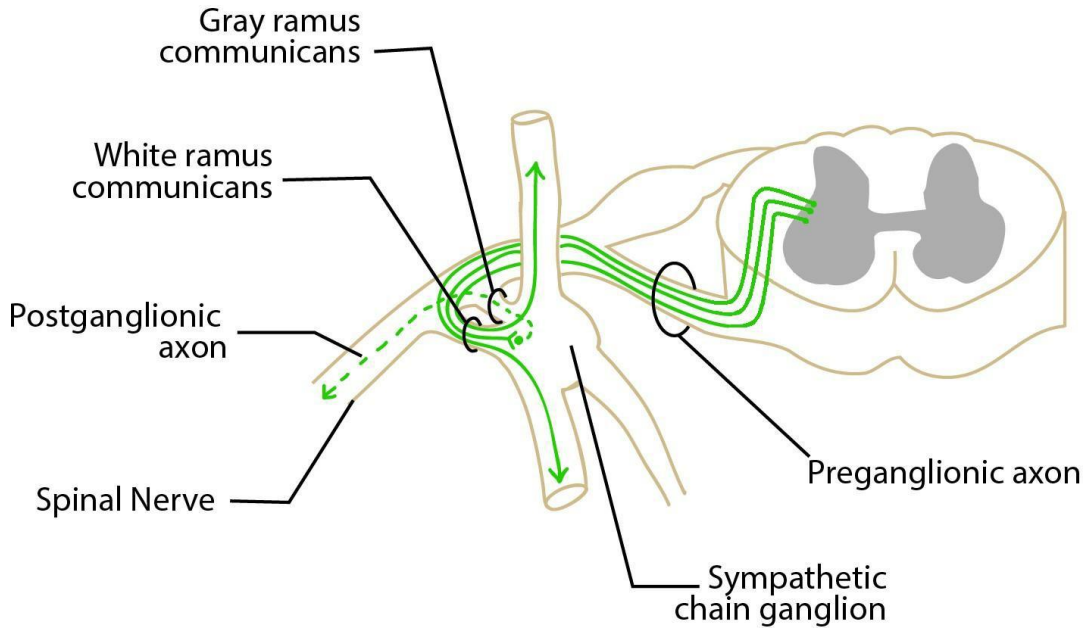


PARAVERTEBRAL(PREAORTIC) GANGLIA

- **Prevertebral ganglia** are in the plexuses that surround the origins of the main branches of the abdominal aorta (for which they are named), such as the two large *celiac ganglia* that surround the origin of the *celiac trunk* (a major artery arising from the aorta).

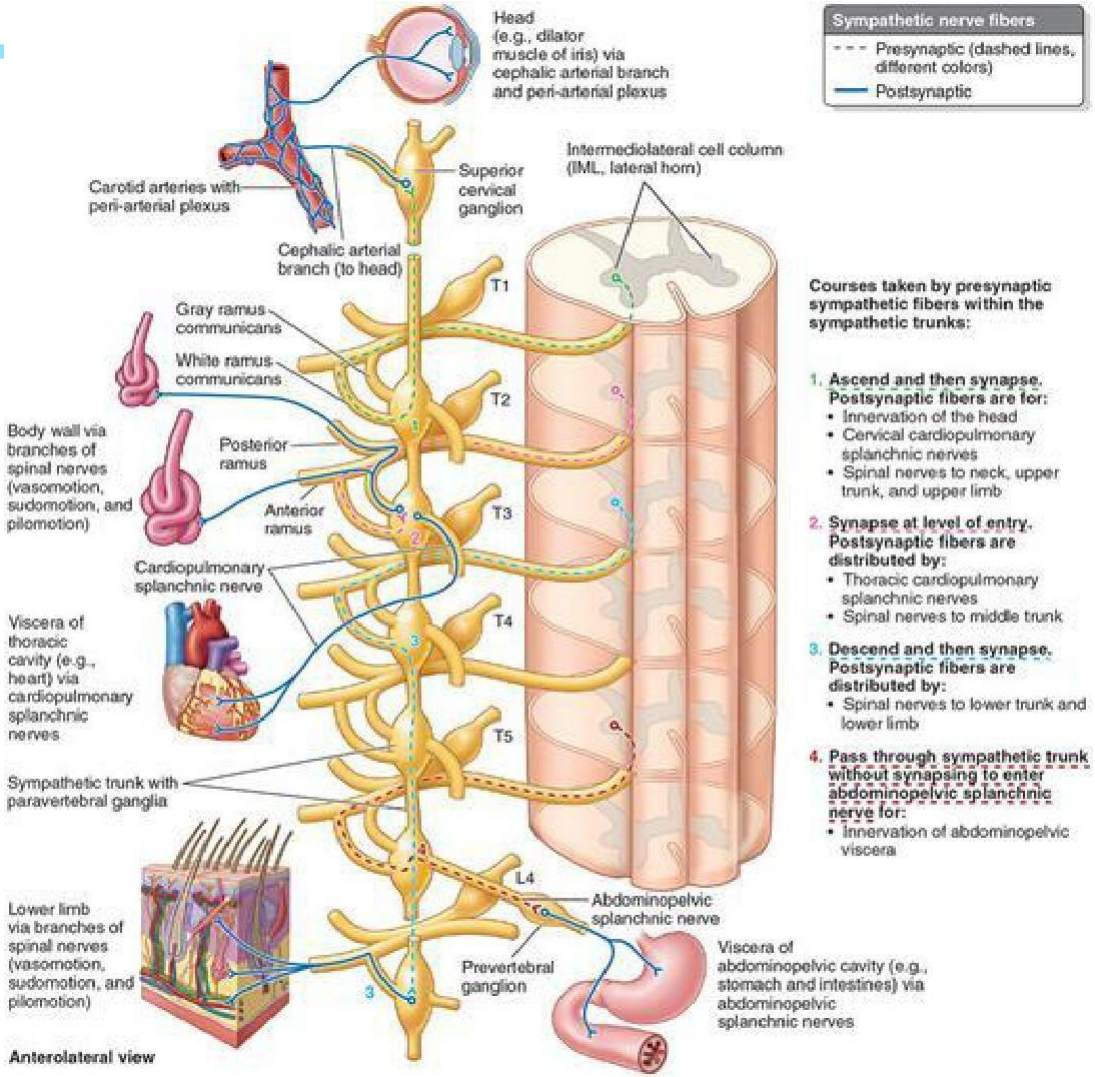


- The axons of presynaptic neurons leave the spinal cord through anterior roots and enter the anterior rami of spinal nerves T1–L2 or L3 (Figs. 1.45 and 1.46).
- Almost immediately after entering, all the presynaptic sympathetic fibers leave the anterior rami of these spinal nerves and pass to the sympathetic trunks through **white rami communicantes** (*communicating branches*).



COURSE TAKEN BY SYMPATHETIC FIBERS

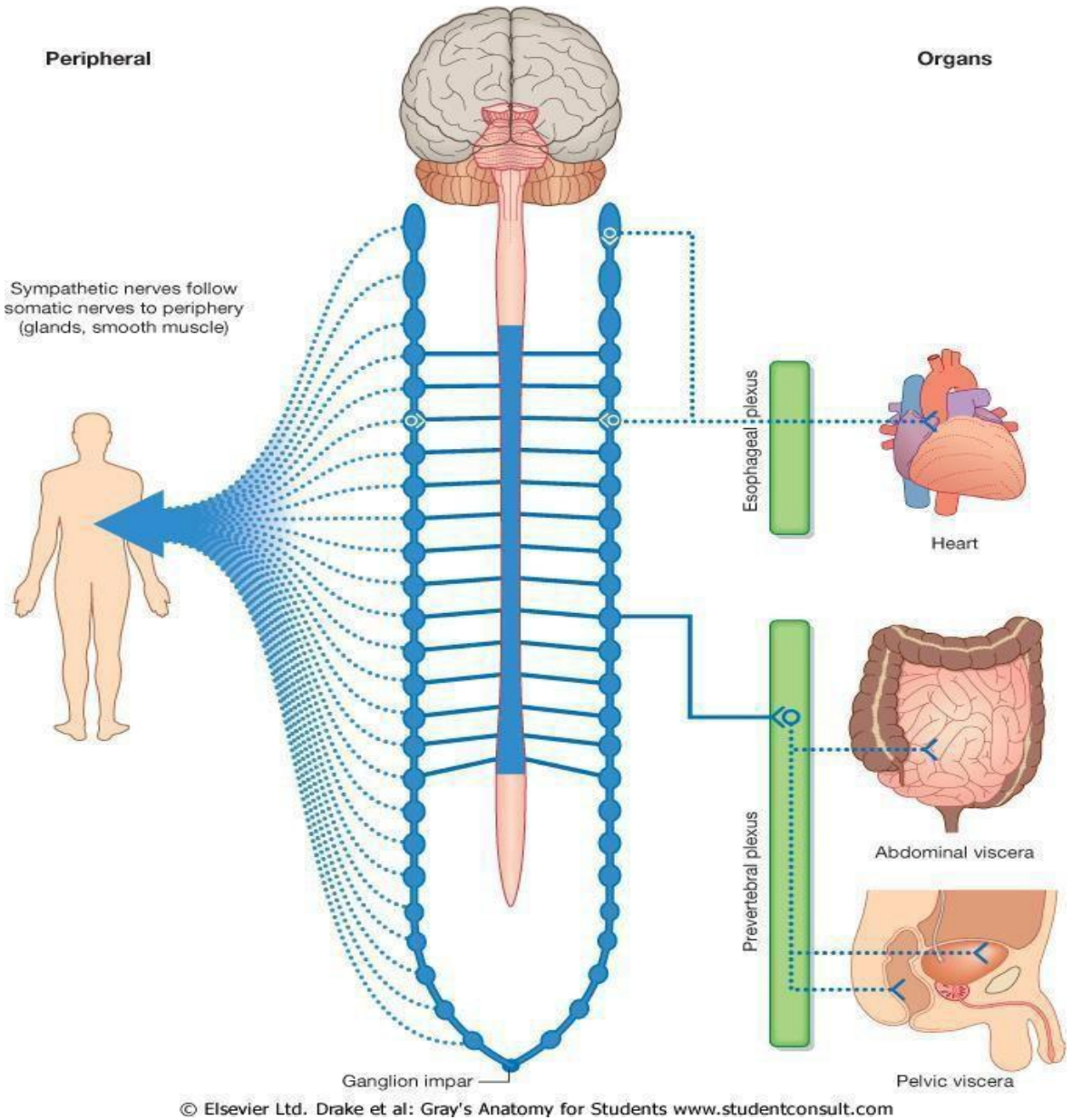
1. **Ascend** in the sympathetic trunk to synapse with a postsynaptic neuron of a higher paravertebral ganglion
2. **Descend** in the sympathetic trunk to synapse with a postsynaptic neuron of a lower paravertebral ganglion
3. **Enter and synapse immediately** with a postsynaptic neuron of the paravertebral ganglion at that level
4. **Pass through the sympathetic trunk without synapsing,** continuing through an abdominopelvic splanchnic nerve to reach the prevertebral ganglia.



SYMPATHETIC NERVOUS SYSTEM

- Those destined for parietal distribution within the neck, body wall, and limbs pass from the sympathetic trunks to adjacent anterior rami of all spinal nerves through **gray rami cummunicantes** and subsequently to all branches of all spinal nerves.

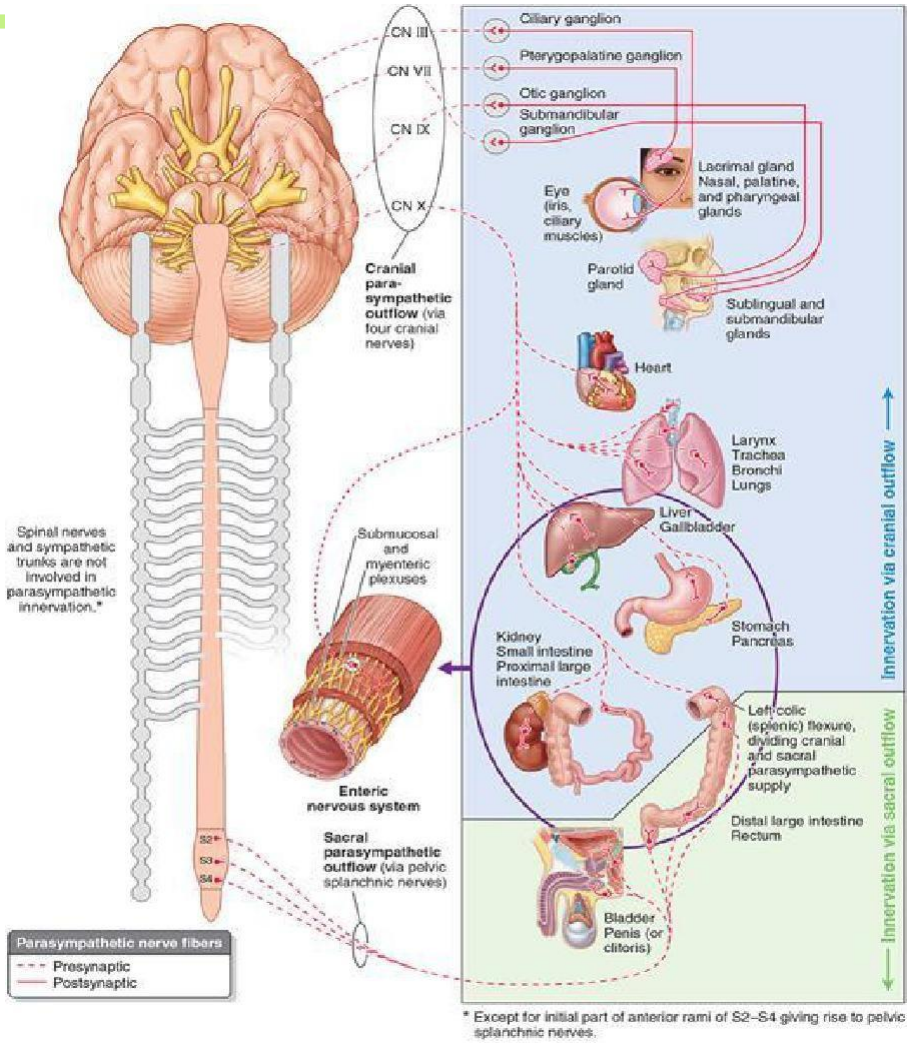
- ***postsynaptic sympathetic fibers are components of virtually all branches of all spinal nerves. By this means and via periarterial plexuses, they extend to and innervate all the body's blood vessels (the sympathetic system's primary function) as well as sweat glands, arrector pilli muscle of the hair.***



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PARASYMPATHETIC DIVISION OF ANS

- ***Presynaptic parasympathetic nerve cell bodies are located in two sites;***
- **In the gray matter of the brainstem**, the fibers exit the CNS within cranial nerves III, VII, IX, and X; these fibers constitute the **cranial parasympathetic outflow**.
- **In the gray matter of the sacral segments of the spinal cord (S2–S4),** the fibers exit the CNS through the sacral spinal nerves S2–S4; these fibers constitute the **sacral parasympathetic outflow**.



PARASYMPATHETIC NERVOUS SYSTEM

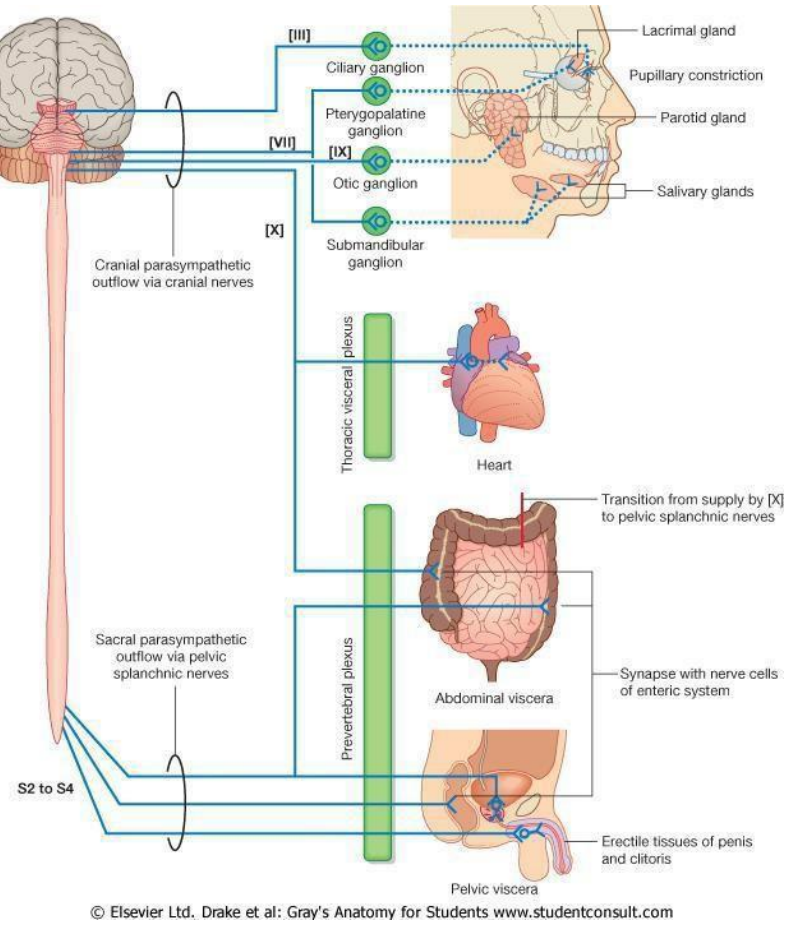
- **The cranial outflow** provides parasympathetic innervation of the head (CN III, VII and IX) and thoracic and abdominal viscera, the cranial outflow through the vagus nerve (CN X).
- **The sacral outflow** to the GI tract supplies only the descending and sigmoid colon and rectum.

PARASYMPATHETIC GANGLIA

Four discrete pairs of parasympathetic ganglia occur in the head.

Elsewhere, presynaptic parasympathetic fibers synapse with postsynaptic cell bodies, which are located in or on the wall of the target organ **(intramural**).

Consequently, in this division, most presynaptic fibers are very long, extending from the CNS to the effector organ, whereas the postsynaptic fibers are very short, running from a ganglion located near or embedded in the effector organ.



REFERENCES

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2. Gray's Anatomy for Students, by Drake, et al, 4th ed., 2019, Elsevier
3. Atlas of Human Anatomy, by F. Netter, 8th ed., 2014, Saunders (Elsevier)